

VIGO AP, Spain

WMO: 080450

Lat: 42.2387N

Lon: 8.6237W

Elev: 261

StdP: 98.23

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.4	2.7	-3.9	2.8	5.9	-2.0	3.3	5.7	11.2	11.9	9.9	11.7	2.4	230	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.8	30.9	20.4	28.9	19.7	27.1	19.0	21.6	28.8	20.6	26.9	19.7	25.2	3.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.4	24.0	18.4	13.7	22.8	17.8	13.2	21.9	64.0	29.0	60.3	27.0	57.3	25.3	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	7.6	6.7	-0.5	34.5	1.4	2.3	-1.5	36.2	-2.2	37.5	-3.0	38.8	-4.0	40.5		
			-1.6	23.4	1.4	1.6	-2.6	24.5	-3.4	25.4	-4.2	26.3	-5.3	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	9.0	9.6	11.7	12.8	15.0	18.0	19.7	20.1	18.5	15.7	11.4	9.6	(6)
(7)		DBStd	4.89	2.44	2.59	3.08	3.34	3.25	3.20	2.94	2.78	2.81	2.94	2.69	2.63	(7)
(8)		HDD10.0	164	49	34	16	8	1	0	0	0	1	15	40	(8)	
(9)		HDD18.3	1694	291	243	205	169	114	44	17	11	31	91	207	272	(9)
(10)		CDD10.0	1731	17	24	69	92	156	241	299	313	256	179	58	27	(10)
(11)		CDD18.3	219	0	0	1	3	12	34	58	65	37	10	0	0	(11)
(12)		CDH23.3	1609	0	0	7	26	93	291	454	474	218	46	0	0	(12)
(13)		CDH26.7	485	0	0	0	2	18	91	151	164	52	7	0	0	(13)
(14)	Wind	WSAvg	3.0	3.1	3.0	3.2	3.2	3.1	3.0	2.9	2.9	2.6	2.8	2.8	3.1	(14)
(15)	Precipitation	PrecAvg	1501	186	118	121	133	108	46	34	46	96	212	214	186	(15)
(16)		PrecMax	2108	364	326	621	395	269	143	92	137	287	367	486	583	(16)
(17)		PrecMin	949	21	0	0	44	19	0	0	5	11	0	11	8	(17)
(18)		PrecStd	290	106	88	125	94	67	36	27	34	65	97	126	124	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.6	19.7	24.1	26.2	28.9	32.1	33.5	34.0	31.0	27.4	20.8	17.3	(19)
(20)			MCWB	11.5	12.8	14.4	16.4	18.0	20.7	21.7	22.1	19.9	17.7	15.4	12.7	(20)
(21)		2%	DB	14.7	16.8	21.4	23.4	26.0	29.3	30.3	30.6	28.0	24.2	18.1	15.3	(21)
(22)			MCWB	12.1	11.5	13.1	15.0	17.2	19.5	20.7	20.3	18.9	17.1	14.8	13.1	(22)
(23)		5%	DB	13.6	14.9	19.0	20.8	23.4	26.9	28.0	28.2	25.9	21.9	16.6	14.3	(23)
(24)			MCWB	11.9	10.7	12.4	14.0	16.5	18.7	20.4	19.6	18.2	16.3	14.2	12.6	(24)
(25)		10%	DB	12.8	13.3	16.6	18.4	20.9	24.2	25.8	25.8	23.7	19.9	15.2	13.4	(25)
(26)			MCWB	11.3	10.6	11.4	13.0	15.3	17.6	19.3	18.9	17.7	15.9	13.3	12.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	13.9	15.6	17.2	20.1	22.2	23.2	23.6	21.1	19.8	17.3	15.1	(27)
(28)			MCDWB	14.8	16.8	21.9	23.8	26.6	29.6	30.9	31.7	28.4	23.8	18.4	15.6	(28)
(29)		2%	WB	13.2	12.9	14.1	15.7	18.2	20.5	21.6	21.5	19.8	18.5	16.0	14.1	(29)
(30)			MCDWB	13.8	14.6	18.8	21.7	23.9	27.3	28.9	28.3	25.8	21.5	17.4	14.7	(30)
(31)		5%	WB	12.4	12.1	13.1	14.6	16.9	19.3	20.5	20.4	18.9	17.6	14.8	13.3	(31)
(32)			MCDWB	13.1	13.7	16.6	19.4	22.2	25.4	26.8	25.9	23.8	20.1	15.9	13.9	(32)
(33)		10%	WB	11.6	11.3	12.3	13.5	15.7	18.1	19.5	19.5	18.1	16.8	13.8	12.3	(33)
(34)			MCDWB	12.4	12.8	15.3	17.4	19.9	23.4	25.0	24.2	22.3	19.0	14.9	13.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.3	7.4	8.1	8.2	8.5	9.1	9.5	9.8	9.2	7.7	6.6	6.3	(35)
(36)			MCDBR	7.1	9.7	12.3	12.3	12.6	13.9	14.0	14.2	12.7	11.0	8.2	6.6	(36)
(37)			MCWBR	5.1	5.7	6.5	6.6	6.6	7.0	6.7	6.3	5.7	5.6	5.0	4.9	(37)
(38)		5% WB	MCDBR	5.0	7.2	9.8	10.7	11.2	12.4	12.6	12.0	10.6	8.3	6.0	5.1	(38)
(39)			MCWBR	4.4	5.1	5.8	6.1	6.2	6.6	6.3	5.9	5.4	4.9	4.6	4.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.326	0.333	0.366	0.366	0.372	0.374	0.364	0.371	0.361	0.357	0.332	0.321	(40)	
(41)		taud	2.504	2.470	2.384	2.392	2.397	2.406	2.432	2.410	2.448	2.482	2.508	2.513	(41)	
(42)		Ebn at Noon	819	872	878	903	902	897	903	887	872	832	803	792	(42)	
(43)		Edh at Noon	73	88	107	114	117	116	112	111	99	86	72	67	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.62	2.61	3.71	4.97	6.03	6.68	6.73	6.09	4.76	2.99	1.89	1.47	(44)	
(45)		RadStd	0.28	0.41	0.60	0.56	0.58	0.46	0.40	0.43	0.32	0.41	0.33	0.23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+34

Nomenclature: See separate page